Capstone Project Weekly Report

Date: August 23, 2024
Report Owner: Sai Prashanth
Team Members: Arvind, Hariharan, Vani, Vedika, Sai Prashanth
Sponsor Company: Deloitte
Project Title: Air Conditioning Demand Forecasting & Supply Chain Optimization

Note: All fields in this form are required.

Project Milestones

Member	Assigned Responsibility	Primary Deliverable / Milestone
Sai Prashanth	Report Owner & Governance	Overall progress tracking, sponsor query aggregation, and executive report synthesis.
Hariharan	Demand & Target Analysis	Evaluation of order/demand dynamics, IDU vs. ODU splits, and target variable definition.
Vani	Data Quality & Integrity Audit	Missing-value analysis, schema anomaly detection, and sales return quantification.
Vedika	SKU-Level EDA & Intermittency	Syntetos-Boylan demand classification, distributional profiling, and top-volume SKU analysis.
Arvind	Architecture & Pipeline Specs	Data dictionary creation, data type mapping, and ETL pipeline normalization rules.

Progress Made in Current Week

During Week 1, the team established a clean data foundation, evaluated target metrics, performed exploratory data analysis (EDA), diagnosed quality anomalies, and categorized demand intermittency.

1. Key Data Portfolio Metrics

- **Dataset Analyzed:**
- **Total Transaction Records:** 82,033 clean billing records spanning July 16, 2022 to June 30, 2024 (625 unique transaction calendar days across 3 fiscal years).
- **Net Volume:** 902,391.0 units (Gross shipments: 909,451.5 units; Sales returns: -7,060.5 units).
- **Active SKU Portfolio:** 154 Room Air Conditioner (RA) SKUs across 5 Regional Hubs (MAA - Chennai: 35.82%, SBD - South Depot: 27.07%, SBD1: 16.78%, COK - Cochin: 11.07%, BLR - Bengaluru: 9.26%).

2. Schema Documentation & Data Quality Audit

- **Schema Mapping:** Analyzed 23 raw fields. Identified 3 redundant system copy columns (`Material.1`, `SALES OFFICE CODE.1`, `Tonnage.1`) and 1 static constant (`SBU/MatGroup3`) targeted for pruning.

- **Structural Temporal Missingness:** Identified 23,603 rows (28.77% of dataset) with null entries for `Billing Doc.`, `Item`, `Stor. Location`, and `Dealer Name`. 100% of these nulls concentrate strictly between April 3, 2024 and June 30, 2024 (Q1 FY24-25), indicating a structural pipeline shift rather than random missingness.

- **Sales Returns & Negative Adjustments:** 1,735 transactions record negative quantities totaling -7,060.5 units (0.78% return rate), primarily in top-volume inverter models (13S/15S) in MAA, BLR, and SBD.

- **Zero Volumes & Formatting:** Identified 243 rows with 0.0 ODU/IDU quantities and minor text variations in `Star Rating`.

3. Demand Dynamics & Target Variable Definition

- **Component Asymmetry (ODU vs. IDU):** Outdoor Units (RA-ODU: 729,297.0 net units / 80.8%) outnumber Indoor Units (RA-IDU: 173,094.0 net units / 19.2%), showing unbundled SAP billing.

- **Target Formulation:** Primary target variable $y_{i,j,t}$ defined for SKU i, depot j, and week t:

$$y_{i,j,t} = \text{Billing Quantity ODU}_{i,j,t} + \text{Billing Quantity IDU}_{i,j,t}$$

- **Syntetos-Boylan Intermittency Classification ($ADI = 1.32$, $CV^2 = 0.49$):**

  - **Lumpy Demand (70.1% / 108 SKUs):** High intermittency ($ADI \geq 1.32$) and high volatility ($CV^2 \geq 0.49$).

  - **Intermittent Demand (18.8% / 29 SKUs):** Sparse demand ($ADI \geq 1.32$) with stable order sizes ($CV^2 < 0.49$).

  - **Erratic Demand (11.0% / 17 SKUs):** Frequent orders ($ADI < 1.32$) with quantity spikes ($CV^2 \geq 0.49$).

  - **Smooth Demand (0.0% / 0 SKUs):** No continuous stationary series exists at daily resolution.

Tasks to Finish in Next Week

1. **Data Preprocessing & Pipeline Build:** Implement automated ETL scripts to drop duplicate columns (`Material.1`, `SALES OFFICE CODE.1`, `Tonnage.1`), handle zero-quantity rows, and standardize star rating text formats.

2. **Feature Engineering:** Construct temporal feature sets including lag variables, rolling seasonal statistics, cooling degree day (CDD) proxies, and hierarchical aggregate keys.

3. **Baseline Modeling Benchmark:** Develop baseline models (Croston SBA, Naive, Moving Average) and benchmark against Zero-Inflated Gradient Boosting regressors (LightGBM/XGBoost) at the weekly level.

Updates/MoM from Sponsor and Faculty Mentor

Updates/MoM from Sponsor

The team formally submitted four strategic technical queries to Deloitte project sponsors to finalize Phase 2 business logic:

1. **Q1 FY24-25 Data Ingestion Shift:** Clarifying whether the 23,603 un-invoiced records (April–June 2024) represent order pipeline off-take, secondary dealer sales, or an SAP export format shift.
2. **Q2 Component Billing Asymmetry:** Confirming if 4:1 ODU-to-IDU ratio stems from separate billing entities or requires independent forecasting models per stream.
3. **Q3 Sales Return Credit Processing:** Determining whether the 1,735 negative return records (-7,060.5 units) should be back-dated to original invoice dates or posted in event weeks.
4. **Q4 Model Granularity & Hierarchy Alignment:** Defining whether inventory targets will deploy at the Weekly SKU-Depot level (154×5 hubs) or National SBU level.

Updates/MoM from Faculty Mentor

- Reviewed Week 1 EDA deliverables, Syntetos-Boylan intermittency categorization, and schema findings.
- Approved the weekly target aggregation level ($y_{i,j,t}$) and recommended benchmarking Croston SBA against tree-based ensemble models for lumpy SKUs.

Challenges

Technical Challenges

- **Severe Demand Intermittency:** 70.1% of SKUs (108 SKUs) exhibit lumpy demand profiles with 0% smooth SKUs, causing standard linear and ARIMA models to fail.
- **Structural Metadata Missingness:** 28.77% of dataset records (23,603 rows) in Q1 FY24-25 lack invoice metadata (`Billing Doc.`, `Dealer Name`, `Item`, `Stor. Location`).
- **Component Billing Imbalance:** Unbundled 4:1 dispatch ratio between ODU (80.8%) and IDU (19.2%) requires careful target variable alignment.

Non-Technical Challenges

- Awaiting formal sponsor feedback from Deloitte on business logic rules for un-invoiced pipeline data and credit return handling.

Mention Any Other Queries/Challenges Regarding the Project to Highlight

Key Highlighted Action Item

Confirmation of production hierarchy alignment (Weekly SKU-Regional Sales Office level vs. Aggregated National SBU level) is required from Deloitte prior to scaling Phase 2 feature engineering pipelines.